

Foundations of Vision transformer

[Vision transformer]

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

1.0 Content Block

Applications ViT have been used in many computer vision tasks with excellent results and in some cases even state-of-the-art, such as in image classification, object detection, video deepfake detection, image segmentation, anomaly detection, image synthesis, cluster analysis, autonomous driving. ViT had been used for image generation as backbones for GAN and for diffusion models (diffusion transformer, or DiT). DINO has been demonstrated to learn useful representations for clustering images and exploring morphological profiles on biological datasets, such as images generated with the Cell Painting assay. In 2024, a 113 billion-parameter ViT model was designed (the largest ViT to date) for weather and climate prediction, and trained on the Frontier supercomputer with a throughput of 1.6 exaFLOPs.

2.0 Content Block

ViT-VQGAN In ViT-VQGAN, there are two ViT encoders and a discriminator. One encodes 8x8 patches of an image into a list of vectors, one for each patch. The vectors can only come from a discrete set of "codebook", as in vector quantization. Another encodes the quantized vectors back to image patches. The training objective attempts to make the reconstruction image (the output image) faithful to the input image. The discriminator (usually a convolutional network, but other networks are allowed) attempts to decide if an image is an original real image, or a reconstructed image by the ViT. The idea is essentially the same as vector quantized variational autoencoder (VQVAE) plus generative adversarial network (GAN). After such a ViT-VQGAN is trained, it can be used to code an arbitrary image into a list of symbols, and code an arbitrary list of symbols into an image. The list of symbols can be used to train into a standard autoregressive transformer (like GPT), for autoregressively generating an image. Further, one can take a list of caption-image pairs, convert the images into strings of symbols, and train a standard GPT-style transformer. Then at test time, one can just give an image caption, and have it autoregressively generate the image. This is the structure of Google Parti.

3.0 Content Block

Medically Supervised MAE Medically Supervised MAE seeks to address the application of MAE's high mask ratios when applied to medical lesion datasets and uses a supervised training set to create local attention maps for medical images in order to constrain which patches are masked out. Medically Supervised MAE achieved state-of-the-art performance as of Jan. 2025 on the classification of medical lesions on the Messidor-2, BTMD, HAM10000, DeepLesion, and ChestXRay2017 datasets Gray Level Co-occurrence Matrix MAE (GLCM-MAE) GCLM-MAE uses GCLM to extract texture information from images in order to preserve texture information. It addresses an issue in which a classic MAE oversmooths images, causing a loss of granular detail that may be important in medical contexts. GLCM-MAE achieves state-of-the-art performance on the identification of gallbladder cancer, breast cancer imaged from ultrasound, pneumonia imaged from X-rays, and COVID-19 imaged from computed tomography as of Jul. 2025. Region-aware MAE, R-MAE R-MAE replaces patch-generating step in the original MAE with an algorithm for assigning individual pixels to regions of interest in an image, which are masked out together. The region encoding architecture is standalone, but can be combined with the MAE for region reconstruction. Siamese MAEs (SiamMAE) SiamMAE is a network designed to apply MAEs to video data.

Samples two frames from a video (compared to one in the original MAE), and labels them as "past" and "future." The network masks out a majority of the patches (~95%) in the future frame, leaves the past frame untouched, and passes both through the MAE encoder block. The decoder architecture is replaced with attention blocks that map patches from the past frame to the future frame for reconstruction. SiamMAE achieves competitive performance against larger models on segmentation and propagation in videos. A similar architecture was BERT ViT (BEiT), published concurrently.

4.0 Content Block

DINO Like the Masked Autoencoder, the DINO (self-distillation with no labels) method is a way to train a ViT by self-supervision. DINO is a form of teacher-student self-distillation. In DINO, the student is the model itself, and the teacher is an exponential average of the student's past states. The method is similar to previous works like momentum contrast and bootstrap your own latent (BYOL). The loss function used in DINO is the cross-entropy loss between the output of the teacher network (f_{θ_t}) and the output of the student network (f_{θ_t}). The teacher network is an exponentially decaying average of the student network's past parameters: $\theta_t = \alpha \theta_t + \alpha (1 - \alpha) \theta_{t-1} + \dots$. The inputs to the networks are two different crops of the same image, represented as $T(x)$ and $T'(x)$, where x is the original image. The loss function is written as $L(f_{\theta_t}(T(x)), f_{\theta_t}(T'(x)))$. One issue is that the network can "collapse" by always outputting the same value (y), regardless of the input. To prevent this collapse, DINO employs two strategies:

5.0 Content Block

A space attention layer is a self-attention layer where each query patch $q_{x,y,t}$ attends to only the key and value patches $k_{x',y',t'}, v_{x',y',t'}$ such that $t = t'$. A time attention layer is where the requirement is $x' = x, y' = y$ instead. The TimeSformer also considered other attention layer designs, such as the "height attention layer" where the requirement is $x' = x, t' = t$. However, they found empirically that the best design interleaves one space attention layer and one time attention layer.